Asset pipeline and UGC

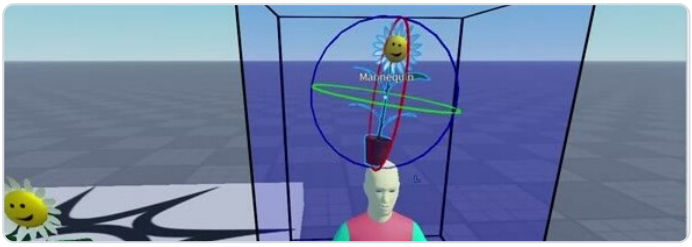
Technical artist, UGC creator

`scenario-roblox-assets`

Turns AI-generated or Blender meshes into Roblox assets at true stud size, facing -Z, under budget and watertight, then previews, audits and validates them locally without uploading.



Live playtest, Studio 0.743: the prepared image-to-3D hat (3,686 triangles, watertight) worn front and side.



Accessory Fitting Tool box around a hat on the mannequin. KatsBits, rigid hat tutorial, 09:32.

USE IT WHEN

- Importing a GLB, FBX or OBJ, including image-to-3D output, into Studio
- Wrong scale, mesh faces backward, too many triangles, or blurry textures
- Preparing a hat or rigid accessory that must pass UGC validation

WHAT THE EXPERTS INSIST ON

- Roblox judges an accessory by its bounding box; rotating off-axis grows the box, so re-center and orient in Blender.
  KatsBits (game-art tutorial publisher)
- Most out-of-bounds errors come from pivot and scale mismatches, not from triangle counts. Joemam's UGC (hobbyist)
- Never use Make Double Sided to hide holes: it costs performance and validation flags the holes later. Roblox Creator Docs (Importer)
- Raw AI meshes need remeshing, pivot and scale work; image-first generation gives more reliable shapes than text.
  CallMeDogey (creator, sponsored video)
- Test every item on several bodies with a run animation; an idle wear test misses what moves. CG Noob (UGC creator)

WORKFLOW, WITH THE GATE BEFORE EACH NEXT STAGE

1. **Triage the GLB. Gate:** triangles, size, map sizes, doubleSided and skins written per asset
2. **Fix in Blender. Gate:** every gate passes and the front view shows the front
3. **Read live rules. Gate:** triangle cap, texture cap and attachment box recorded from Studio
4. **Preview and audit in Studio. Gate:** size within 0.01 stud of Blender, triangle count matches, no audit error
5. **Wear test the accessory. Gate:** attachment error under 0.01 stud; the crown clears the head
6. **Validate locally. Gate:** item valid with no string; every single-fault copy fails
7. **Hand off the import. Gate:** imported MeshPart size equals the preview; the audit runs again

KEY NUMBERS

Triangles per rigid or layered accessory	**4,000; one mesh in a place: 20,000**
Texture density	**256 px per 2 x 2 x 2 studs, power of two**
Avatar ColorMap, live MaxTextureSize	**1024 px; other avatar maps 256**
Hat box with no scale type	**3 x 4 x 3 studs; ProportionsNormal 1.87 x 2.5 x 1.87**
Hat crown opening (wear test)	**1.35 studs for a 1.16-stud default R15 head**
Local validator rate limit	**10 calls per 60 s**

CHECKS THE AGENT RUNS ON ITS OWN WORK

- Blender report: every gate passes, and the raw input fails as a negative control
- AssetAudit returns no error; Size and triangle count equal the Blender report
- Validator leaves no string fixable without an upload; single-fault copies all fail
- Contact sheet: fronts face -Z, props on the ground, hat seated front and side

HANDOFFS

Receives GLB or FBX meshes from scenario-3d, one-mesh PBR sets from scenario-textures, concept images from scenario-game-assets.

Delivers MeshParts at true size with bottom pivot and gate report to environment; rigged imports to animation; batches to performance.

TOOLKIT RBX_ASSETS.PY

- `glb_info()`: triangles, size, maps and flags of a GLB, offline
- `blender_prep()`: headless Blender pass for scale, facing, pivot, budget, holes
- `gates()`: pass or fail each prop or accessory rule

MAIN EXPERT SOURCES

KatsBits
Game-art tutorial publisher, Accessory Fitting Tool, rigid hat (2026)

Joemam's UGC
Hobbyist UGC creator, Fix UGC size boundaries (2025)

CG Noob
UGC creator and add-on author, Layered clothing in Blender (2024)

Roblox Creator Docs
Official documentation, Importer, rigid accessory specifications, validation system

STATUS

Tests 12 offline checks pass

In app 12 live tests pass in Roblox Studio 0.743; nothing uploaded, UGC tested up to Roblox validation only

G9 UGC and asset pipeline

without skills	7/15
with skills	14/15